



Set C

1. Valid Parentheses

Given a string s containing just the characters '(', ')', '{', '}', '[', and ']', determine if the input string is valid.

An input string is valid if:

Open brackets must be closed by the same type of brackets.

Open brackets must be closed in the correct order.

Every close bracket has a corresponding open bracket of the same type.

1.1. Examples:

Input: $s = "()"$

Output: true

Input: $s = "()[]\{\}"$

Output: true

Input: $s = "()["$

Output: false

Input: $s = "([])"$

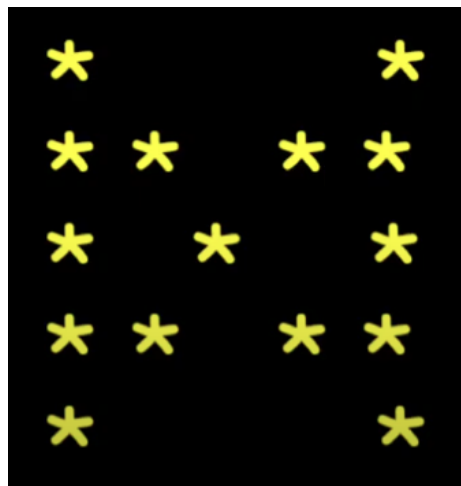
Output: true

1.2. Constraints:

1.2.1. $1 \leq s.length \leq 104$

1.2.2. s consists of parentheses only '()[]\{\}'.

2. Draw the below pattern with dynamic input rows (the below example is when the input is 5):





3. Maximum Difference Between Increasing Elements

Given a 0-indexed integer array `nums` of size `n`, find the maximum difference between `nums[i]` and `nums[j]` (i.e., `nums[j] - nums[i]`), such that $0 \leq i < j < n$ and `nums[i] < nums[j]`.

Return the maximum difference. If no such `i` and `j` exists, return `-1`.

Note: Need to solve this in single loop with $O(n)$ time.

3.1. Examples:

Input: `nums = [7,1,5,4]`

Output: 4

Explanation: The maximum difference occurs with `i = 1` and `j = 2`, `nums[j] - nums[i] = 5 - 1 = 4`.

Note that with `i = 1` and `j = 0`, the difference `nums[j] - nums[i] = 7 - 1 = 6`, but `i > j`, so it is not valid.

Input: `nums = [1,5,2,10]`

Output: 9

Explanation:

The maximum difference occurs with `i = 0` and `j = 3`, `nums[j] - nums[i] = 10 - 1 = 9`.

Input: `nums = [9,4,3,2]`

Output: -1

Explanation:

There is no `i` and `j` such that `i < j` and `nums[i] < nums[j]`.

3.2. Constraints:

3.2.1. `n == nums.length`

3.2.2. $2 \leq n \leq 1000$

3.2.3. $1 \leq \text{nums}[i] \leq 10^9$



Instructions (Read Carefully): -

1. Time Duration: **3 hours.**
2. Write Programs in a dynamic and interactive way and use OOPs wherever necessary.
3. All the inputs must be taken from the User and it should be dynamic. Do not use static input/values.
4. Do not use any built-in functions/methods in any of the programs. Create your own functions/methods.
5. You do not have permission to access Google.
6. You don't have permission to access Social Networking sites.
7. You can't download anything and can't copy-paste code.
8. You can't use ChatGPT.
9. Keep your phone on silent and in your bag.
10. A few guidelines to make your code acceptable and JavaScript special instructions: [☰ Interview Guidelines](#)